

ARM-450 — URDF Reference

Robot name arm450 · 6 revolute joints · 450 mm chain · generated 2026-08-20

This document covers the URDF alone: the kinematic chain, every link and joint, the verification it has passed, and the complete source. Kinematics theory (Modified-DH, transforms, FK derivation) is in ARM450_KINEMATICS.pdf; how to run it is in ARM450_RUN.pdf.

1. The chain

```
world
├─ base_link          fixed at origin
│   └─ link1          joint1    Z roll    z = 50    base yaw / turret
│       └─ link2      joint2    Y pitch   z = 40    shoulder
│           └─ link3   joint3    Y pitch   z = 119   elbow
│               └─ link4 joint4    Z roll    z = 119   forearm roll
│                   └─ link5 joint5    Y pitch   z = 62    wrist pitch
│                       └─ link6 joint6    Z roll    z = 30    tool roll
│                           └─ tcp        fixed    z = 30    tool centre point
│                               └─ tool      fixed    z = 0     fitted end effector
│                                   └─ tool_tip fixed    z = 72    working point
```

Eleven links, ten joints, six of them actuated. world, tcp and tool_tip are massless frames — world gives RViz a fixed root to attach to, tcp is the point the IK solves for, and tool_tip is where the fitted tool actually works from.

The tool link

Added 2026-08-22. Before that the URDF stopped at link6 and RViz drew a bare J6 face, even though the sine path had been re-verified collision-free with a gripper fitted and the workspace recomputed for it. All of that lived in assemble.py — the CAD and collision assembly — and none of it reached the model RViz loads.

tool is a fixed link on link6 carrying the quick-change adapter, the tool body and (for the gripper) both jaws as a single combined mesh, placed by assemble.build() so the thing RViz draws is the geometry the collision check cleared.

fitted	mass	tool_tip at	note
gripper	74.6 g	z = 72 mm from link6	42 mm past the J6 face
dock probe	62.4 g	z = 66 mm from link6	36 mm past the J6 face
none	—	—	bare face, the old behaviour

Switch with `EMIT=1 TOOL=dock python3 generate_urdf_meshes.py` — see ARM450_RUN.pdf §9A.

The tool is payload, not arm. The mass check in preflight stage 5 sums the arm links only and reports the tool separately; adding it in would have inflated the total by 75 g and quietly broken the one check that ties the model to the 1450 g budget.

The 450 mm budget

base	50	
shoulder rise	40	
L2 upper arm	119	
L3 forearm	119	
J4 → J5	62	
J5 → J6	30	
J6 → TCP	30	

TOTAL	450 mm	✓ requirement met exactly

generate_urdf.py asserts this sum at build time, so the chain cannot silently drift past 450 mm. It also asserts $L2 == LINK_L$ and $L4_SPLIT == LINK_L$ against the CAD parameters, which is what keeps the URDF and the printed parts describing the same robot.

$L2 = L3 = 119$ mm is deliberate. The inner radius of a reachable annulus is $|L2 - L3|$, so equal links drive the dead zone to zero. A 60 000-pose FK sweep confirms no unreachable inner region.

2. Joints

joint	type	parent → child	origin z (mm)	axis	range	effort (N·m)
world_to_base	fixed	world → base_link	0	—	—	—
joint1	revolute	base_link → link1	50	Z (roll)	$\pm 165^\circ$	2.94
joint2	revolute	link1 → link2	40	Y (pitch)	$\pm 115^\circ$	4.9
joint3	revolute	link2 → link3	119	Y (pitch)	$\pm 150^\circ$	2.94
joint4	revolute	link3 → link4	119	Z (roll)	$\pm 165^\circ$	2.94
joint5	revolute	link4 → link5	62	Y (pitch)	$\pm 110^\circ$	2.94
joint6	revolute	link5 → link6	30	Z (roll)	$\pm 175^\circ$	2.94
link6_to_tcp	fixed	link6 → tcp	30	—	—	—

Axis convention: at $q = 0$ every joint is at its home position and the arm stands straight up along +Z. joint2 carries the ST3250 (4.9 N·m) because the shoulder sees the largest gravity moment; every other joint is an ST3215 at 2.94 N·m.

Why the limits are asymmetric in usefulness: joint5 at $\pm 110^\circ$ is the tightest, and it is the joint that runs out of travel first when tracing on a horizontal surface. Vertical board tracing — the Rsine task — stays well inside it.

3. Links

link	mass (kg)	ixx (kg·m ²)	iyy	izz
world	frame only			
base_link	0.1800	1.412e-04	1.412e-04	2.074e-04
link1	0.1000	3.417e-05	2.034e-05	2.784e-05
link2	0.1450	2.013e-04	1.813e-04	4.037e-05
link3	0.1300	1.805e-04	1.625e-04	3.619e-05
link4	0.0600	3.172e-05	2.343e-05	1.671e-05
link5	0.0600	1.700e-05	8.700e-06	1.671e-05
link6	0.0900	2.550e-05	1.306e-05	2.506e-05
tcp	frame only			
total	0.765			

Link mass totals 0.765 kg. This is the URDF's dynamic model, not the build mass — it excludes servos, bearings and shafts, which the mass gate accounts for separately (1144 g all-in against the 1.2 kg limit).

Inertias are solid-cuboid approximations about each link's own centre, computed from the real clamshell section (50.0 × 29.0 mm). They are good enough for visualisation and gravity-torque estimates. They are not measured, so do not use them for high-bandwidth control tuning without re-deriving from the CAD.

4. Geometry

Visual and collision geometry are boxes and a cylinder sized to the real printed section, not meshes. That is a deliberate choice: primitives keep collision checks cheap and make the kinematics legible.

STL meshes for all seven bodies exist in `arm450_description/meshes/` and can be swapped into the `<visual>` blocks when a photorealistic render is wanted. Doing so changes nothing kinematic.

material	rgba	used for
steel	0.62 0.70 0.79 1	base, link2, link3, link5
accent	0.18 0.49 0.60 1	link1, link4
servo	0.78 0.34 0.24 1	link6

5. Verification

check	result
check_urdf parse	Passed — single tree, root world, 9 links, no orphans
Chain sums to 450 mm	450.0 mm, asserted at generation

check	result
URDF ↔ CAD link length	119.0 == 119.0, asserted
FK vs Modified-DH, 5000 random poses	0.000 nm position error
Dead zone (60 000-pose sweep)	0 mm — no unreachable inner region
robot_state_publisher load	Publishes full TF tree, all 6 joints
Sine trajectory replay	240 waypoints, 0.193 mm rms path error

The DH rotation differs from the URDF by a constant 180° about Z. That is frame labelling only — position agreement is exact, and the constant offset is documented in ARM450_KINEMATICS.pdf.

6. How it is generated

The URDF is generated, never hand-edited. `generate_urdf.py` imports the section and link length straight from `cad/params.py`, so a change to the printed geometry propagates to the model automatically.

```
cd ~/ros2_ws/arm450_design
python3 generate_urdf.py          # writes arm450.urdf
cp arm450.urdf ../src/arm450_description/urdf/
```

If you edit `arm450.urdf` by hand, the next generation overwrites it. Change `generate_urdf.py` or `cad/params.py` instead.

7. Complete source

```
<?xml version="1.0"?>
<!-- ARM-450 : 6-DOF arm, 450 mm overall, L2 = L3 = 145 mm.
      Generated by generate_urdf.py – do not hand-edit. -->
<robot name="arm450">
  <material name="steel"><color rgba="0.62 0.70 0.79 1"/></material>
  <material name="accent"><color rgba="0.18 0.49 0.60 1"/></material>
  <material name="servo"><color rgba="0.78 0.34 0.24 1"/></material>
  <link name="world"/>
  <joint name="world_to_base" type="fixed">
    <parent link="world"/><child link="base_link"/>
    <origin rpy="0 0 0" xyz="0 0 0"/>
  </joint>
  <link name="base_link">
    <inertial>
      <origin rpy="0 0 0" xyz="0 0 0.025000"/>
      <mass value="0.1800"/>
      <inertia ixx="0.00014118" ixy="0" ixz="0" iyy="0.00014118" iyz="0" izz="0.00020736"/>
    </inertial>
    <visual>
      <origin rpy="0 0 0" xyz="0 0 0.025000"/>
      <geometry><cylinder radius="0.04800" length="0.05000"/></geometry>
      <material name="steel"/>
    </visual>
    <collision>
```

```

    <origin rpy="0 0 0" xyz="0 0 0.025000"/>
    <geometry><cylinder radius="0.04800" length="0.05000"/></geometry>
  </collision>
</link>
<joint name="joint1" type="revolute">
  <parent link="base_link"/>
  <child link="link1"/>
  <origin rpy="0 0 0" xyz="0 0 0.050000"/>
  <axis xyz="0 0 1"/>
  <limit effort="2.94" velocity="4.6"
    lower="-2.879793" upper="2.879793"/>
  <dynamics damping="0.05" friction="0.02"/>
</joint>
<link name="link1">
  <inertial>
    <origin rpy="0 0 0" xyz="0 0 0.020000"/>
    <mass value="0.1000"/>
    <inertia ixx="0.00003417" ixy="0" ixz="0" iyy="0.00002034" iyz="0" izz="0.00002784"/>
  </inertial>
  <visual>
    <origin rpy="0 0 0" xyz="0 0 0.020000"/>
    <geometry><box size="0.02900 0.05000 0.04000"/></geometry>
    <material name="accent"/>
  </visual>
  <collision>
    <origin rpy="0 0 0" xyz="0 0 0.020000"/>
    <geometry><box size="0.02900 0.05000 0.04000"/></geometry>
  </collision>
</link>
<joint name="joint2" type="revolute">
  <parent link="link1"/>
  <child link="link2"/>
  <origin rpy="0 0 0" xyz="0 0 0.040000"/>
  <axis xyz="0 1 0"/>
  <limit effort="4.9" velocity="3.7"
    lower="-2.007129" upper="2.007129"/>
  <dynamics damping="0.05" friction="0.02"/>
</joint>
<link name="link2">
  <inertial>
    <origin rpy="0 0 0" xyz="0 0 0.059500"/>
    <mass value="0.1450"/>
    <inertia ixx="0.00020132" ixy="0" ixz="0" iyy="0.00018127" iyz="0" izz="0.00004037"/>
  </inertial>
  <visual>
    <origin rpy="0 0 0" xyz="0 0 0.059500"/>
    <geometry><box size="0.02900 0.05000 0.11900"/></geometry>
    <material name="steel"/>
  </visual>
  <collision>
    <origin rpy="0 0 0" xyz="0 0 0.059500"/>
    <geometry><box size="0.02900 0.05000 0.11900"/></geometry>
  </collision>
</link>

```

```

    </collision>
</link>
<joint name="joint3" type="revolute">
  <parent link="link2"/>
  <child link="link3"/>
  <origin rpy="0 0 0" xyz="0 0 0.119000"/>
  <axis xyz="0 1 0"/>
  <limit effort="2.94" velocity="4.6"
    lower="-2.617994" upper="2.617994"/>
  <dynamics damping="0.05" friction="0.02"/>
</joint>
<link name="link3">
  <inertial>
    <origin rpy="0 0 0" xyz="0 0 0.059500"/>
    <mass value="0.1300"/>
    <inertia ixx="0.00018049" ixy="0" ixz="0" iyy="0.00016252" iyz="0" izz="0.00003619"/>
  </inertial>
  <visual>
    <origin rpy="0 0 0" xyz="0 0 0.059500"/>
    <geometry><box size="0.02900 0.05000 0.11900"/></geometry>
    <material name="steel"/>
  </visual>
  <collision>
    <origin rpy="0 0 0" xyz="0 0 0.059500"/>
    <geometry><box size="0.02900 0.05000 0.11900"/></geometry>
  </collision>
</link>
<joint name="joint4" type="revolute">
  <parent link="link3"/>
  <child link="link4"/>
  <origin rpy="0 0 0" xyz="0 0 0.119000"/>
  <axis xyz="0 0 1"/>
  <limit effort="2.94" velocity="4.6"
    lower="-2.879793" upper="2.879793"/>
  <dynamics damping="0.05" friction="0.02"/>
</joint>
<link name="link4">
  <inertial>
    <origin rpy="0 0 0" xyz="0 0 0.031000"/>
    <mass value="0.0600"/>
    <inertia ixx="0.00003172" ixy="0" ixz="0" iyy="0.00002343" iyz="0" izz="0.00001671"/>
  </inertial>
  <visual>
    <origin rpy="0 0 0" xyz="0 0 0.031000"/>
    <geometry><box size="0.02900 0.05000 0.06200"/></geometry>
    <material name="accent"/>
  </visual>
  <collision>
    <origin rpy="0 0 0" xyz="0 0 0.031000"/>
    <geometry><box size="0.02900 0.05000 0.06200"/></geometry>
  </collision>
</link>

```

```

<joint name="joint5" type="revolute">
  <parent link="link4"/>
  <child link="link5"/>
  <origin rpy="0 0 0" xyz="0 0 0.062000"/>
  <axis xyz="0 1 0"/>
  <limit effort="2.94" velocity="4.6"
    lower="-1.919862" upper="1.919862"/>
  <dynamics damping="0.05" friction="0.02"/>
</joint>
<link name="link5">
  <inertial>
    <origin rpy="0 0 0" xyz="0 0 0.015000"/>
    <mass value="0.0600"/>
    <inertia ixx="0.00001700" ixy="0" ixz="0" iyy="0.00000870" iyz="0" izz="0.00001671"/>
  </inertial>
  <visual>
    <origin rpy="0 0 0" xyz="0 0 0.015000"/>
    <geometry><box size="0.02900 0.05000 0.03000"/></geometry>
    <material name="steel"/>
  </visual>
  <collision>
    <origin rpy="0 0 0" xyz="0 0 0.015000"/>
    <geometry><box size="0.02900 0.05000 0.03000"/></geometry>
  </collision>
</link>
<joint name="joint6" type="revolute">
  <parent link="link5"/>
  <child link="link6"/>
  <origin rpy="0 0 0" xyz="0 0 0.030000"/>
  <axis xyz="0 0 1"/>
  <limit effort="2.94" velocity="4.6"
    lower="-3.054326" upper="3.054326"/>
  <dynamics damping="0.05" friction="0.02"/>
</joint>
<link name="link6">
  <inertial>
    <origin rpy="0 0 0" xyz="0 0 0.015000"/>
    <mass value="0.0900"/>
    <inertia ixx="0.00002550" ixy="0" ixz="0" iyy="0.00001306" iyz="0" izz="0.00002506"/>
  </inertial>
  <visual>
    <origin rpy="0 0 0" xyz="0 0 0.015000"/>
    <geometry><box size="0.02900 0.05000 0.03000"/></geometry>
    <material name="servo"/>
  </visual>
  <collision>
    <origin rpy="0 0 0" xyz="0 0 0.015000"/>
    <geometry><box size="0.02900 0.05000 0.03000"/></geometry>
  </collision>
</link>
<joint name="link6_to_tcp" type="fixed">
  <parent link="link6"/><child link="tcp"/>

```

```
    <origin rpy="0 0 0" xyz="0 0 0.030000"/>
  </joint>
  <link name="tcp"/>
</robot>
```